

Surface Tension

8.1 Introduction

If a sewing needle is placed carefully on a water surface, it makes a small depression on the surface and rests there without sinking. The needle does not sink though its density is higher than that of water. In a glass-tube with a small bore (capillary tube) is dipped in water, the water rises in the capillary tube. The level of water inside the capillary tube is higher than the level of water outside it. Rain drops are spherical in shape. Similarly a liquid flowing out of a dropper, comes out in the form of fine drops and not as a continuous stream. Water in trees and planes rises upwards against the force of gravity. All similar phenomena are due to the existence of a boundary surface between the liquid and other substances.

Mutual attraction between the molecules of the same substances is called *cohesion*.

Mutual attraction between the molecules of different substances is called *adhesion*. These forces are effective only when the distance between the neighbouring molecules is extremely small i.e. of the order of 10^{-7} cm. The greatest distance at which the molecules can attract each other is called *molecular range*. A sphere having a radius equal to the molecular range, with the molecule as the centre, is called the *sphere of influence*. The molecule is attracted by all other molecules within the sphere of influence.

8.2 Surface Tension

Surface tension is the property of a liquid by which the free surface of a liquid behaves like a stretched membrane.

Consider an imaginary line AB dividing the liquid surface into two parts. Since there are forces of attraction between the molecules of the liquid, the stresses act on this imaginary line (Fig. 8.1).

The molecules on one side pull away the molecules on the other side. Therefore, if the liquid was actually separated into two parts with AB as the boundary line, work would have to be done. Hence, production of a large amount of a free surface requires an increase in the potential energy of the liquid. The undisturbed liquid tries to minimize its potential energy. It must take a shape in which its *free surface area is as small as possible*. Its surface tends to contract and *force per unit length acting on either side of the imaginary line drawn on the liquid surface at rest is called the surface tension*. The direction of the force is tangential to the surface and perpendicular to the line.

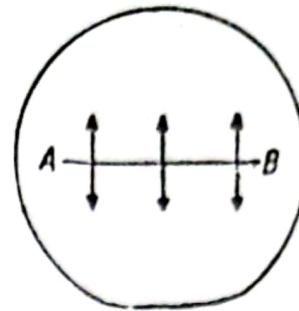


Fig. 8.1

$$\text{Surface Tension} = \frac{\text{Force}}{\text{Length}}$$

The unit of surface tension in CGS system is dynes per centimetre and in rationalised MKS system is newtons per metre.

The dimensions of surface tension are $[MT^{-2}]$.

8.3 Explanation of Surface Tension

Consider a molecule A of a liquid lying well within the free surface of a liquid. It has a sphere of influence of the order 10^{-7} cm. This molecule is attracted by all the molecules lying within the sphere of influence (Fig. 8.2). The resultant force acting on the molecule A due to all the molecules is zero.

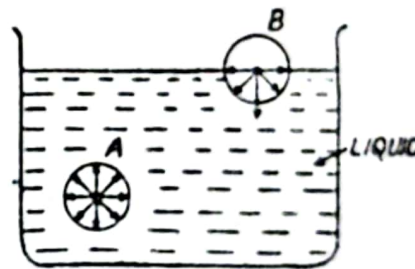


Fig. 8.2

Consider a molecule B on the surface of the liquid (Fig. 8.2). This molecule experiences forces of attraction only due to the molecules lying in the lower half of the sphere of influence. The resultant of all these forces is in the downward direction. Due to this, the net inward force on the molecules lying on the surface of the liquid, the surface is stretched. Therefore, the free surface of a liquid at rest behaves like a stretched membrane.

8.4 Examples of Surface Tension

1. Take a metal ring. Dip it in soap solution. Place a moistened loop of cotton thread over the film (Fig. 8.3). The loop has an irregular shape.

Touch the soap film within the loop with a hot needle. The thread takes a circular shape (Fig. 8.4). Every point on the thread experiences a force radially outwards and parallel to the film.

2. Place a pin or sewing needle gently on the surface of water. It floats. There is depression on the surface of water and the

surface behaves like a stretched membrane. Press the needle with the finger. The layer due to surface tension breaks and the needle sinks in water.

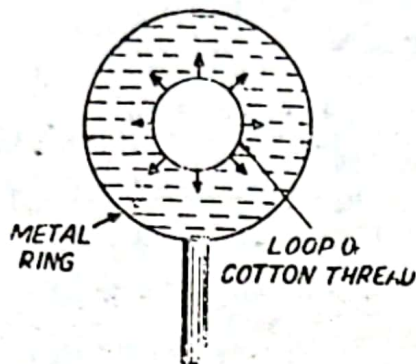


Fig. 8.3

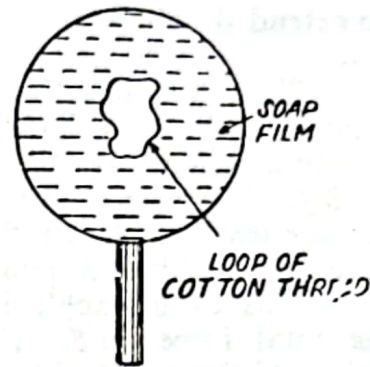


Fig. 8.4

3. Take a shaving brush. Dip it in water. Take the brush out. The hairs of the brush cling together due to the thin layers of water.

4. Take a mixture of spirit and water such that the density of the mixture is equal to the density of olive oil.

With the help of a pipette, leave a drop of olive oil inside the mixture of spirit and water. The olive oil drop assumes a spherical shape. (Fig. 8.5).

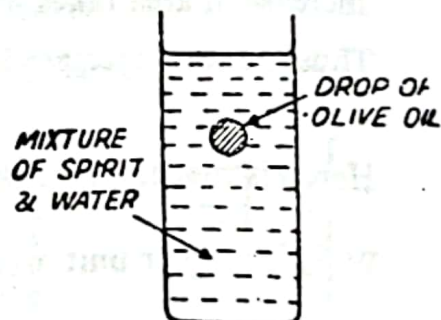


Fig. 8.5

Due to surface tension, the tendency of a liquid surface is to keep its area minimum for a given volume. *For a given volume the surface area of a sphere is minimum.*

5. Falling rain drops are spherical in shape due to surface tension.

6. Leads shots are manufactured in factories based on the property of surface tension. Molten lead is allowed to fall through the end of a narrow tube. The lead drops assume spherical shape due to surface tension.

7. Mosquitoes breed on the surface of free water. The layer of water due to surface tension supports them. Oil has lesser surface tension than water. When a drop of oil is sprayed on the surface of water, due to higher surface tension of water, the oil is stretched in all directions as a thin film. The surface tension of oil surface is small and mosquitoes cannot breed on it. *Therefore to prevent the breeding of mosquitoes in rainy season, oil is sprayed on the surface of water in pools and ponds.*

8.5 Surface Energy and Surface Tension

As discussed already, a liquid surface contracts spontaneously. Free energy is associated with it and work must be done to extend the surface.

Take a rectangular frame $PQRS$ with a horizontal wire AB . (Fig. 8.6). Dip it in a soap solution. A film is formed across $ABRQ$. The wire AB is pulled up due to surface tension of the film. To keep the wire in equilibrium and in its position, a force has to be applied downwards. Let the total force be F . Here F is equal to $w_1 + w_2$, where w_1 is the weight of the wire and w_2 is the extra weight to be used.

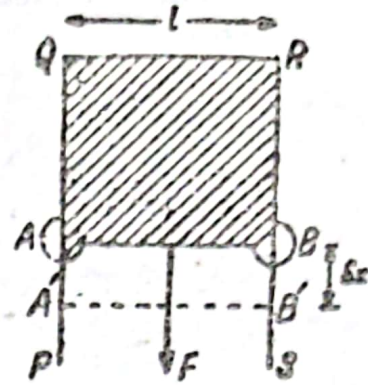


Fig. 8.6

It is found that the same force F will hold the wire at rest in any position, regardless of the area of the surface.

When the wire AB is pulled downwards through a distance δx , the area of the film is increased. The molecules in the bulk of the liquid move into the surface layers.

It is assumed that the temperature remains constant.

Work done $= F \times \delta x$.

Increase in area takes place on both the sides of the film.

Thus the total increase in the surface area of the liquid film

$$= 2(l \times \delta x)$$

Here l is the length of the wire AB

$$\text{Work done per unit area} = \frac{F \times \delta x}{2l \times \delta x} = \frac{F}{2l}$$

This work done per unit area gives the increase in potential energy per unit area of the film. This increase in energy per unit area is called the surface energy.

Also the force $F = T \times 2l$

Here T is the surface tension.

$$\therefore T = \frac{F}{2l}$$

Therefore, numerically surface tension is equal to surface energy.

It was assumed that the temperature remains constant throughout the process. But, actually, the film gets cooled when it is stretched. Therefore extra energy has to be supplied to restore the original temperature of the film.

Therefore, the surface energy is equal to the sum of the mechanical work done and the heat absorbed from the atmosphere.

Suppose H is the quantity of heat absorbed per unit area from the atmosphere. (Here H is in units of work). Then,

$$S \times 2l \delta x = 2lT \times \delta x + H \times 2l \delta x$$

or

$$S = T + H \quad \dots(i)$$

Here S is the surface energy, T the surface tension and H is the amount of heat absorbed per unit area from the atmosphere.

From equation (i)

$$T = S - H \quad \dots(ii)$$

$(S - H)$ is called the mechanical part of the surface energy and is also called the free energy of the surface.

Therefore, the surface tension is equal to the mechanical part $(S - H)$ of the surface energy and is also defined as the amount of work done in increasing the surface area of the film by one unit under isothermal conditions.

If the process is adiabatic $H = 0$

From equations (ii)

$$T = S \quad \dots(iii)$$

The units of free surface energy in CGS system is ergs per sq cm and in MKS system is joules per sq metre.

8.6 Pressure Difference Across a Spherical Surface

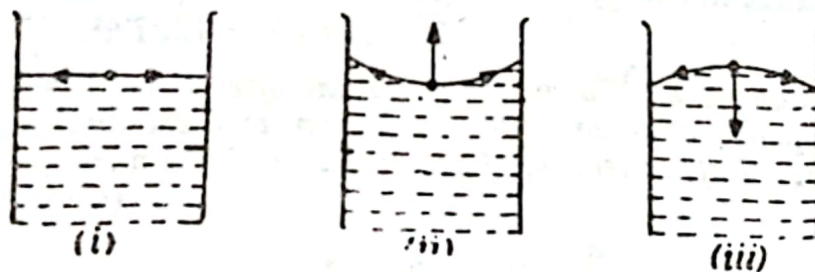


Fig. 8.7

A liquid surface can be plane, concave or convex as shown in Fig. 8.7

(1) When the liquid surface is plane, the resultant force of surface tension on a molecule is zero.

(2) When the liquid surface is convex, the resultant force is outward.

(3) When the liquid surface is concave, the resultant force is inward.

Thus for curved liquid surface the, pressure on the concave side is greater than on the convex side. The difference of pressure depends upon the surface tension of the liquid and the radius of curvature of its surface.

Example 8.1. A liquid drop of radius R breaks up into 64 small drops. Calculate the change in energy.

Let the surface tension of the liquid be T .

Radius of the bigger drop $= R$

Surface area of the bigger drop $= 4\pi R^2$

Volume of the bigger drop $= \frac{4}{3} \pi R^3$

Let the radius of each small drop be r .

$$\therefore \frac{4}{3} \pi R^3 = 64 \times \frac{4}{3} \pi r^3$$

$$\therefore R = 4r$$

$$r = \frac{R}{4}$$

Surface area of each small drop
 $= 4\pi r^2$

$$= 4\pi \left(\frac{R}{4} \right)^2 = \frac{\pi R^2}{4}$$

Surface area of 64 small drops

$$= \frac{64 \times \pi R^2}{4} = 16\pi R^2$$

Increase in surface area

$$= 16\pi R^2 - 4\pi R^2 = 12\pi R^2$$

Change in energy $= (\text{surface tension}) (\text{increase in area})$

$$= T \times (12\pi R^2) = 12\pi TR^2.$$

Example 8.2. Calculate the amount of energy needed to break a drop of petrol of volume 10^{-6} m^3 into a thousand million drops of equal size. Surface tension of petrol is $26 \times 10^{-3} \text{ N/m}$.

(Gorakhpur 1977)

$$\frac{4}{3} \pi R^3 = 10^{-6} \text{ m}^3$$

$$R^3 = \frac{3 \times 10^{-6}}{4\pi}$$

$$R = \left(\frac{3}{4\pi} \right)^{1/3} \times 10^{-2} \text{ m}$$

Also $\frac{4}{3} \pi R^3 = 10^9 \times \frac{4}{3} \pi r^3$

$$10^{-6} = 10^9 \times \frac{4}{3} \pi r^3$$

$$r = \left(\frac{3}{4\pi} \right)^{1/3} \times 10^{-5} \text{ m}$$

Increase in surface area $= 10^9 \times 4\pi r^2 - 4\pi R^2$

$$A = 4\pi \left[10^9 \left(\frac{3}{4\pi} \right)^{2/3} \times 10^{-10} - \left(\frac{3}{4\pi} \right)^{2/3} \times 10^{-4} \right]$$

$$A = 4\pi \times \left(\frac{3}{4\pi}\right)^{2/3} \times 999 \times 10^{-4} \text{ m}^2$$

$$A = (4\pi)^{1/3} \times (3)^{2/3} \times 999 \times 10^{-4}$$

$$T = 26 \times 10^{-3} \text{ N/m}$$

$$\text{Energy required} = T \times A$$

$$= 26 \times 10^{-3} \times (4\pi)^{1/3} \times (3)^{2/3} \times 999 \times 10^{-4}$$

$$= 0.01163 \text{ J}$$

Exar. ple 8.3. Calculate the work done in spraying a spherical drop of mercury of radius 10^{-3} m into a million drops of equal size. Surface tension of mercury = $550 \times 10^{-3} \text{ N/m}$. (Delhi 1976)

$$R = 10^{-3} \text{ m}$$

$$\frac{4}{3} \pi R^3 = 10^6 \times \frac{4}{3} \pi r^3$$

$$r = \frac{R}{100} = \frac{10^{-3}}{100} = 10^{-5} \text{ m}$$

$$\text{Increase in surface area} = 10^6 \times 4\pi r^2 - 4\pi R^2$$

$$A = 4\pi [10^6 \times (10^{-5})^2 - (10^{-3})^2]$$

$$A = 4\pi \times 99 \times 10^{-6} \text{ m}^2$$

$$\text{Work done} = T \times A$$

$$= 550 \times 10^{-3} \times 4\pi \times 99 \times 10^{-6}$$

$$= 1.6844 \times 10^{-4} \text{ J}$$

Example 8.4. Calculate the amount of energy needed to break a drop of water of diameter $2 \times 10^{-3} \text{ m}$ into 10^9 droplets of equal size. Surface tension of water $72 \times 10^{-3} \text{ N/m}$. (Delhi 1976)

$$R = 10^{-3} \text{ m}$$

$$\frac{4}{3} \pi R^3 = 10^9 \times \frac{4\pi}{3} r^3$$

$$R = 10^3 r$$

$$r = \frac{R}{10^3} = \frac{10^{-3}}{10^3} = 10^{-6} \text{ m}$$

$$\text{Increase in surface area}$$

$$A = 10^9 \times 4\pi r^2 - 4\pi R^2$$

$$A = 4\pi [10^9 \times (10^{-6})^2 - (10^{-3})^2]$$

$$A = 4\pi \times 999 \times 10^{-6} \text{ m}^2$$

$$T = 72 \times 10^{-3} \text{ N/m}$$

$$\text{Energy required} = T \times A$$

$$= 72 \times 10^{-3} \times 4\pi \times 999 \times 10^{-6} \text{ J}$$

$$= 1.9040 \times 10^{-4} \text{ J}$$

Example 8.5. What amount of energy will be liberated if 1000 droplets of water each 10^{-8} m in diameter coalesce to form one large spherical drop. Surface tension of water = 72×10^{-3} N/m.

[Delhi (Hons) 1976]

$$r = 0.5 \times 10^{-8} \text{ m}$$

$$\frac{4}{3} \pi R^3 = 1000 \times \frac{4}{3} \pi r^3$$

$$R = 10r = 5 \times 10^{-8} \text{ m}$$

Decrease in surface area

$$\begin{aligned} &= 1000 \times 4\pi r^2 - 4\pi R^2 \\ &= 4\pi (1000 \times 25 \times 10^{-18} - 25 \times 10^{-16}) \\ &= 4\pi \times 225 \times 10^{-16} \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Energy liberated} &= (72 \times 10^{-3})(4\pi \times 225 \times 10^{-16}) \text{ J} \\ &= 2.035 \times 10^{-14} \text{ J} \end{aligned}$$

Example 8.6. A spherical soap bubble of radius 1 cm is blown in air. How much energy will be needed to increase the radius to 3 cm. Surface tension of soap solution is 0.03 N/m. [Calcutta 1979]

$$\text{Here, } r_1 = 1 \text{ cm} = 0.01 \text{ m}$$

$$r_2 = 3 \text{ cm} = 0.03 \text{ m}$$

$$\begin{aligned} \text{Increase in area} &= 2 [4\pi(r_2^2 - r_1^2)] \\ &= 8\pi[(0.03)^2 - (0.01)^2] \\ &= 8\pi[(0.03)^2 - (0.01)^2] \\ &= 64\pi \times 10^{-4} \text{ m}^2 \end{aligned}$$

$$\text{Surface Tension, } T = 0.03 \text{ N/m}$$

$$\begin{aligned} \text{Energy required} &= \text{Increase in area} \times \text{surface Tension} \\ &= 64\pi \times 10^{-4} \times 0.03 \\ &= 6.033 \times 10^{-4} \text{ J} \end{aligned}$$

Example 8.7 A soap bubble is slowly enlarged from a radius of 0.01 m to 0.1 m. Calculate the work done in the process. Surface tension of soap solution = 26×10^{-3} N/m. (Delhi 78)

$$r_1 = 0.01 \text{ m}$$

$$r_2 = 0.1 \text{ m}$$

$$\begin{aligned} \text{Increase in area} &= 2[4\pi(r_2^2 - r_1^2)] = 8\pi[(0.1)^2 - (0.01)^2] \\ &= 8\pi \times 0.0099 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Surface energy per unit area} \\ &= 26 \times 10^{-3} \text{ J/m}^2 \end{aligned}$$

Increase in the energy of the film

$$= 26 \times 10^{-3} \times 8\pi \times 0.0099$$

$$= 208\pi \times 99 \times 10^{-7} \text{ J}$$

$$= 6.471 \times 10^{-3} \text{ J}$$

Example 8.8. A mercury drop of radius 1 cm is sprayed into one million drops of the same size. Calculate the energy expended. Surface tension of mercury = $35 \times 10^{-3} \text{ N/m}$.

Here $T = 35 \times 10^{-3} \text{ N/m}$

$$R = 1 \text{ cm}$$

$$= 10^{-2} \text{ m}$$

Radius of each small drop = r

$$\therefore \frac{4}{3} \pi R^3 = 10^6 \times \frac{4}{3} \pi r^3$$

$$R = 100 r$$

$$r = \frac{R}{100} = \frac{10^{-2}}{100} = 10^{-4} \text{ m}$$

$$\begin{aligned} \text{Increase in area} &= (10^6 \times 4\pi r^2) - 4\pi R^2 \\ &= [10^6 \times 4\pi (10^{-4})^2] - 4\pi (10^{-2})^2 \\ &= 4\pi [10^{-2} - 10^{-4}] \\ &= 4\pi \times 99 \times 10^{-4} \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Energy expended} &= (\text{surface tension}) (\text{increase in area}) \\ &= 35 \times 10^{-3} \times 4\pi \times 99 \times 10^{-4} \text{ J} \\ &= 4.355 \times 10^{-3} \text{ J} \end{aligned}$$

Example 8.9. An air bubble of radius 0.1 mm is situated just below the surface of water. Calculate the gauge pressure (excess of pressure) inside the air bubble. Surface Tension of water = $7.2 \times 10^{-2} \text{ N/m}$.

Here $r = 0.1 \text{ mm} = 10^{-4} \text{ m}$

$$T = 7.2 \times 10^{-2} \text{ N/m}$$

$$\begin{aligned} \text{Excess of pressure} &= \frac{2T}{r} = \frac{2 \times 7.2 \times 10^{-2}}{10^{-4}} \\ &= 1440 \text{ N/m}^2 \end{aligned}$$

Example 8.10. Calculate the excess of pressure between the inside and outside of a soap bubble of radius 1 cm. Surface tension of soap solution is $3.2 \times 10^{-2} \text{ N/m}$.

Here $r = 1 \text{ cm} = 10^{-2} \text{ m}$

$$T = 3.2 \times 10^{-2} \text{ N/m}$$

$$\begin{aligned} \text{Excess of pressure} &= \frac{4T}{r} = \frac{4 \times 3.2 \times 10^{-2}}{10^{-2}} \\ &= 12.8 \text{ N/m}^2 \end{aligned}$$

8.7 Excess of Pressure Inside a Spherical Liquid Drop or an Air Bubble in a Liquid

Consider a spherical liquid drop of radius r (or an air bubble inside a liquid) (Fig. 8.8). The molecules near the surface of the drop experience a resultant pull inward due to surface tension.

Therefore, the pressure inside the drop must be greater than the pressure outside. This excess of pressure can be calculated and it depends upon the surface tension of the liquid and the radius of the drop.

Imagine the drop to be divided into two equal hemispheres by a horizontal plane $ABCD$. Consider the equilibrium of the upper hemisphere. Let the pressures inside and outside be P_1 and P_2 respectively.

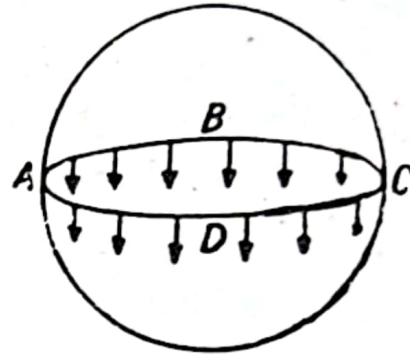


Fig. 8.8

The thrust on the hemisphere due to surface tension + the thrust on the hemisphere due to external pressure P_2 = the thrust on the hemisphere due to the internal pressure P_1 .

$$\text{The face area } ABCD = \pi r^2$$

$$\text{The thrust due to } P_2 = P_2 \times \pi r^2$$

$$\text{The thrust due to } P_1 = P_1 \times \pi r^2$$

$$\begin{aligned} \text{The thrust due to surface tension} \\ = T \times 2\pi r. \end{aligned}$$

The surface tension acts round the circumference of the drop which has a length

$$= 2\pi r.$$

$\therefore$ For equilibrium,

$$T \times 2\pi r + P_2 \times \pi r^2 = P_1 \times \pi r^2$$

$$(P_1 - P_2) = \frac{2T}{r} \quad \dots(i)$$

Let the excess of pressure be p

$$\therefore p = P_1 - P_2$$

$$\therefore p = \frac{2T}{r} \quad \dots(ii)$$

The unit of excess of pressure in C.G.S. system is dynes/sq. cm and in M.K.S. system is newtons/sq metre.

8.8 Excess of Pressure Inside a Soap Bubble

A soap bubble has two liquid surfaces in contact with air, one inside the bubble and the other outside the bubble. Therefore the total length

$$= 2 \times 2\pi r = 4\pi r$$

For equilibrium,

$$T \times 4\pi r + P_2 \times \pi r^2 = P_1 \times \pi r^2$$

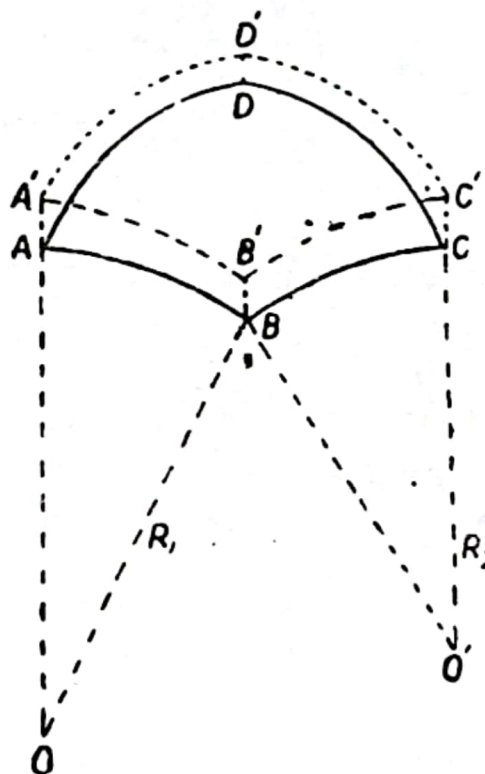
or
$$(P_1 - P_2) = \frac{4T}{r} \quad \dots(i)$$

$\therefore$ Excess of pressure,

$$p = \frac{4T}{r} \quad \dots(ii)$$

Thus, the excess of pressure is inversely proportional to the radius of the drop or the bubble. The pressure of air inside a small bubble is greater than the pressure of air inside a larger bubble.

8.9 Difference of Pressure Across a Curved Surface



$$AB = x, \quad BC = y, \quad BB' = \delta z$$

$$A'B' = x + \delta x, \quad B'C' = y + \delta y$$

Fig. 8.9

If we consider a curved membrane, the element of the membrane is in equilibrium only if the excess of pressure acts on the concave side. This excess of pressure balances the effect due to surface tension. The magnitude of the excess of pressure can be calculated as follows:

Let $ABCD$ be an element of the curved surface having principal radii R_1 and R_2 (Fig. 8.9). Consider that the element has one surface only. Suppose the excess of pressure on the concave side is p .

Surface area of $ABCD = xy$.

Let the surface area xy be given a small displacement δz , so that it takes the position $A'B'C'D'$ parallel to itself.

The surface area increases. The increase in surface area

$$= \delta(xy) = x \cdot \delta y + y \cdot \delta x$$

Let the surface tension be T

$$\therefore \text{Work done} = T \times (x \cdot \delta y + y \cdot \delta x) \quad \dots(i)$$

Also the work done by pressure

$$= p \cdot x \cdot y \cdot \delta z \quad \dots(ii)$$

Equating (i) and (ii)

$$p \cdot x \cdot y \cdot \delta z = T(x \cdot \delta y + y \cdot \delta x) \quad \dots(iii)$$

The triangles ABO and $A'B'O$ are similar,

$$\therefore \frac{A'B'}{OB'} = \frac{B}{OB} = \frac{A'B' - AB}{OB' - OB}$$

$$\frac{x + \delta x}{R_1 + \delta z} = \frac{x}{R_1} = \frac{\delta x}{\delta z}$$

$$\text{or} \quad \delta x = \frac{x}{R_1} \cdot \delta z$$

Similarly it can be proved that

$$\text{or} \quad \delta y = \frac{y}{R_2} \cdot \delta z$$

Substituting the values of δx and δy in equation (iii), we get

$$p \cdot x \cdot y \cdot \delta z = T \left(\frac{xy \delta z}{R_2} + \frac{xy \delta z}{R_1} \right)$$

$$p = T \left(\frac{1}{R_2} + \frac{1}{R_1} \right)$$

$$\text{or} \quad p = T \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad \dots(iv)$$

This expression is also true for a liquid drop or an air bubble in a liquid.

When the centres of curvature of AB and BC are on the opposite sides of the film and it is displaced, one side of the curvilinear rectangle increases and the other decreases. In that case, the excess of pressure

$$p = T \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(v)$$

When the liquid surface has two free surfaces as in the case of a soap film, the excess of pressure,

$$p = 2T \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad \dots(vi)$$

When the centres of curvature are on the opposite sides,

$$p = 2T \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(vii)$$

Let us consider some special cases.

Special Cases :

(1) In the case of a liquid drop or an air bubble,

$$R_1 = R_2 = R$$

From equation (iv),

the excess of pressure,

$$p = T \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{2T}{R} \quad \dots(viii)$$

(2) In the case of a soap bubble, there are two surfaces and

$$R_1 = R_2 = R$$

From equation (vi),

$$p = 2T \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{4T}{R} \quad \dots(ix)$$

(3) If the surface is cylindrical, one of the radii $R_1 = R$ and the other radius $R_2 = \infty$

$$\therefore p = T \left(\frac{1}{R} + \frac{1}{\infty} \right)$$

$$p = \frac{T}{R} \quad \dots(x)$$

For a hollow cylindrical film,

$$p = 2T \left(\frac{1}{R} + \frac{1}{\infty} \right)$$

$$p = \frac{2T}{R} \quad \dots(xi)$$

8.10 Angle of Contact

When a glass plate is dipped in *water*, the water molecules cling to the surface of glass and the water molecules rise along the plate. The shape of water surface is as shown in Fig. 8.10. The angle of contact is θ .

When the glass plate is dipped in *Mercury*, the mercury molecules cling to the surface and the liquid is depressed along the plate as shown in Fig. 8.11.

The angle of contact is θ . Hence, the angle of contact is defined as the angle made by the tangent at the point of contact of the liquid surface with tangent plane to the solid surface inside the liquid.

The angle of contact in the case of water is acute (less than 90°) as shown in Fig. 8.10.

In the case of mercury the angle of contact is obtuse (more than 90°) as shown Fig. 8.11.

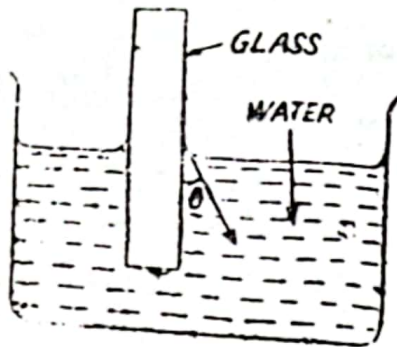


Fig. 8.10

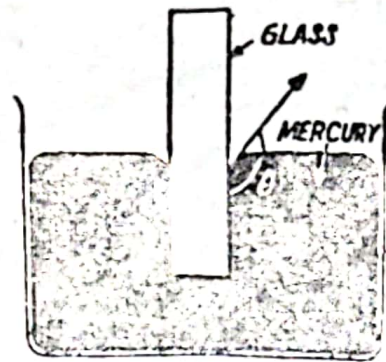


Fig. 8.11

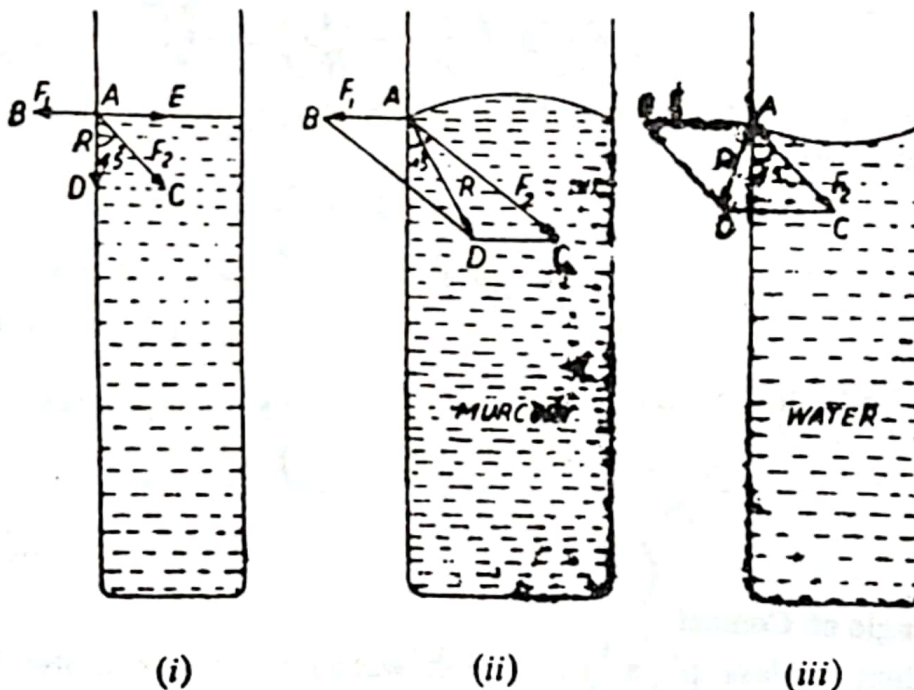
Explanation

Fig. 8.12

Consider a liquid in a glass tube. The molecule at A experiences (i) force of adhesion F_1 between it and the glass molecules (ii) force of cohesion F_2 between it and the molecules of the liquid. F_2 acts along AC making an angle of 45° with the walls of the tube. Let us consider three possible cases.

1. Suppose the surface of the liquid is horizontal as shown in Fig. 8.12 (i). F_2 is the force of cohesion acting along AC . The component of F_2 along AE is $F_2 \sin \theta$. F_1 is the force of adhesion along AB . Here

$$F_1 = F_2 \sin \theta$$

Thus the molecule A does not experience any force in the horizontal direction. The only force is in the vertical direction and is equal to $F_2 \cos \theta$. Here $\theta = 45^\circ$ and the angle of contact is 90° .

2. Suppose the surface of the liquid is convex as shown in Fig. 8.12 (ii). Here F_2 is along AC and $F_2 \sin 45^\circ$ is greater than F_1 . The resultant force due to F_1 and F_2 is R and is along AD . The angle of contact is more than 90° and the liquid surface is convex. The surface of mercury in glass is convex.

3. Suppose the surface of the liquid is concave as shown in Fig. 8.12 (iii). Here F_2 is along AC and $F_2 \sin 45^\circ$ is less than F_1 . The resultant force due to F_1 and F_2 is R and is along AD . The angle of contact is less than 90° and the liquid surface is concave. Water in glass has a concave surface.

In general, liquid that wet the glass surface have a concave surface and the liquids that do not wet the glass surface, have a convex surface.

Also, for liquids having higher force of adhesion with glass, the surface is concave and for liquids having lesser force of adhesion with glass, the surface is convex. Moreover, the angle of contact depends upon the nature of the liquid and the solid in contact with the liquid.

8.11 Determination of Angle of Contact

Angle of Contact for Mercury in Glass

The angle of contact for mercury in glass can be determined with the help of a simple apparatus as shown in Fig. 8.13.

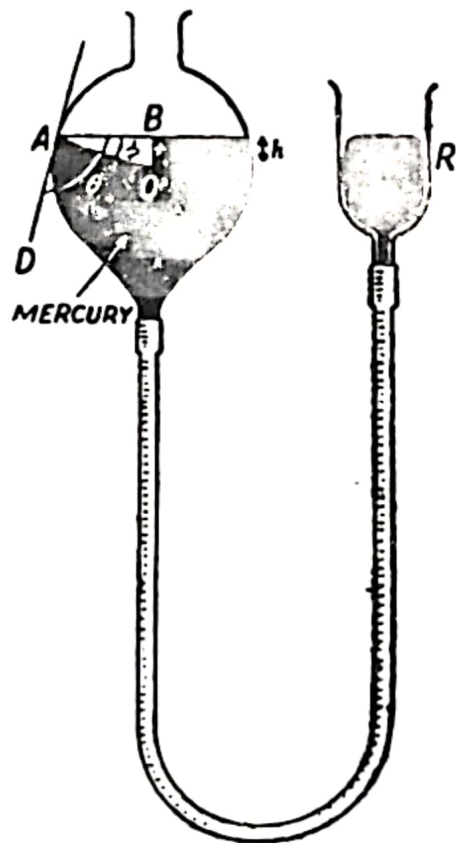


Fig. 8.13

Here

$$h = \frac{2T}{r\rho g}$$

$$T = 70 \times 10^{-3} \text{ N/m}$$

$$r = 4 \times 10^{-4} \text{ m}$$

$$\rho = 10^3 \text{ kg/m}^3$$

$$g = 9.8 \text{ m/s}^2$$

$$h = \frac{2 \times 70 \times 10^{-3}}{4 \times 10^{-4} \times 10^3 \times 9.8}$$

$$h = 3.572 \times 10^{-3} \text{ m}$$

Example 8.21. In a capillary tube water rises to a height of 0.1 m. In the same capillary tube mercury is depressed by $3.42 \times 10^{-2} \text{ m}$.

Angle of contact for water = 0

Angle of contact for mercury = 135°

Calculate the surface tension of mercury given that the surface tension of water is $72 \times 10^{-3} \text{ N/m}$. Density of mercury = $13.6 \times 10^3 \text{ kg/m}^3$.

$$h\rho g = \frac{2T \cos \theta}{r}$$

(i) For water,

$$T_1 = \frac{h_1 \rho_1 g r}{2 \cos \theta_1} \quad \dots (i)$$

(ii) For mercury,

$$T_2 = \frac{h_2 \rho_2 g r}{2 \cos \theta_2} \quad \dots (ii)$$

$$\frac{T_2}{T_1} = \left(\frac{h_2}{h_1} \right) \left(\frac{\rho_2}{\rho_1} \right) \left(\frac{\cos \theta_1}{\cos \theta_2} \right)$$

Here

$$h_2 = -3.42 \times 10^{-2} \text{ m}$$

$$h_1 = 0.1 \text{ m}$$

$$\rho_2 = 13.6 \times 10^3 \text{ kg/m}^3$$

$$\rho_1 = 10^3 \text{ kg/m}^3$$

$$\cos \theta_1 = 1$$

$$\cos \theta_2 = -0.707$$

$$T_1 = 72 \times 10^{-3} \text{ N/m}$$

$$\therefore T_2 = 72 \times 10^{-3} \times \left(\frac{-3.42 \times 10^{-2}}{0.1} \right) \left(\frac{13.6 \times 10^3}{10^3} \right) \left(\frac{1}{-0.707} \right)$$

$$T_2 = 72 \times 10^{-3} \times 0.342 \times 13.6 \times 1.412$$

$$T_2 = 0.4728 \text{ N/m}$$

8.13 Capillarity

When a capillary tube of fine bore is dipped in water, water rises in the tube. [Fig. 8.15]

[A tube with a fine bore is called a capillary tube].

On the other hand if the tube is dipped in mercury, there is depression of mercury level in the tube [Fig. 8.15 (ii)].

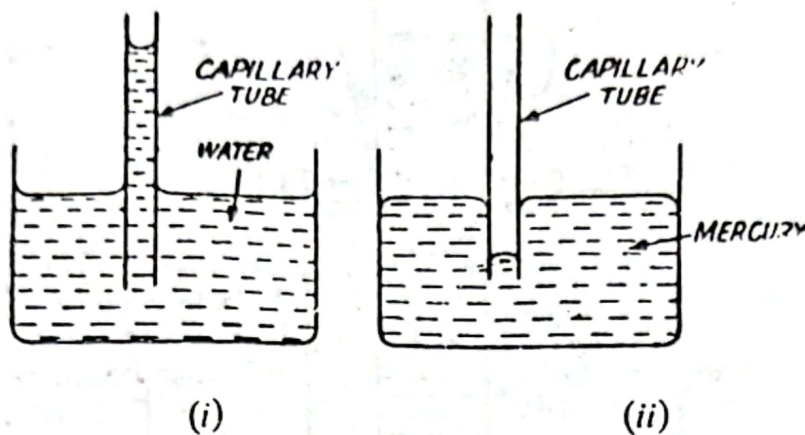


Fig. 8.15

This property of rise or depression of a liquid inside a capillary tube is called capillarity. It is one of the most important effects of surface tension.

In general, the liquids that wet the glass, rise inside the capillary tube while those which do not wet the glass show a depression inside the capillary tube.

8.14 Expression for Surface Tension

Consider a capillary tube of radius r dipped in water. Due to surface tension, the water rises inside the capillary tube. Let h be the height of water in the tube above the free surface of water in the vessel (Fig. 8.16). The meniscus is concave as shown in the figure. At the points A, B etc., the force of surface tension T acts tangential to the surface at the points of contact with the wall of the capillary tube.

This force per unit length T refers to the force experienced by the molecule at A due to

- (i) the force of adhesion due to glass molecules and
- (ii) the force of cohesion due to liquid molecules. Consequently the glass molecules at A, B etc. attract the liquid molecules with a force per unit length T in the directions AP, BQ etc.

Resolve the force per unit length T along AP into two rectangular components

- (i) $T \cos \theta$ acting vertically upwards.
- (ii) $T \sin \theta$ acting in the horizontal direction.

The horizontal components due to all the molecules along the circumference cancel each other, while only the vertical components are effective.

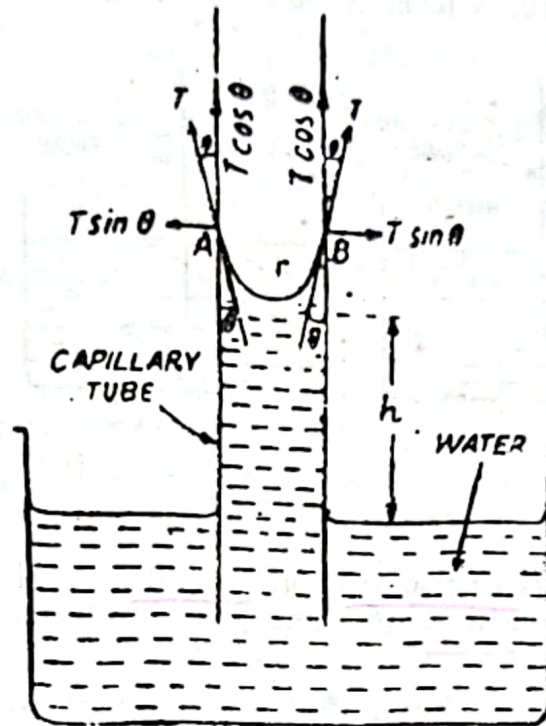


Fig. 8.16

Force per unit length = $T \cos \theta$

Circumference = $2\pi r$

Total upward force = $T \cos \theta \times 2\pi r$... (i)

This upward force balances the weight of the liquid column in the capillary tube.

Height of the liquid column = h

Volume of the liquid in the meniscus

$$= (\pi r^2 \times r - \frac{2}{3}\pi r^3) = \frac{\pi r^3}{3}$$

Volume of liquid in the column

$$= \pi r^2 h + \frac{\pi r^3}{3} = \pi r^2 \left[h + \frac{r}{3} \right]$$

Density of the liquid = ρ

Weight of the liquid = $\pi r^2 \left(h + \frac{r}{3} \right) \rho \cdot g$... (ii)

Equating (i) and (ii)

$$2\pi r T \cos \theta = \pi r^2 \left(h + \frac{r}{3} \right) \rho \cdot g$$

$$T = \frac{r \left(h + \frac{r}{3} \right) \rho \cdot g}{2 \cos \theta} \quad \dots (iii)$$

When the angle of contact is zero,

$$\cos \theta = 1$$

$$T = \frac{r \left(h + \frac{r}{3} \right) \rho \cdot g}{2} \quad \dots(iv)$$

8.15 Determination of Surface Tension of Water

Surface tension of water can be determined experimentally using capillary method. As shown in Fig. 8.16, a capillary tube is dipped in water. The height of the liquid column in the capillary tube is found with the help of a travelling microscope. The internal diameter of the capillary tube is also found with the help of a travelling microscope.

Suppose for a particular tube, the radius is r , the height of water is h and the density of water is ρ . Then the surface tension of water at room temperature,

$$T = \frac{r \left(h + \frac{r}{3} \right) \rho g}{2 \cos \theta}$$

Taking $\theta = 0, \cos 0 = 1$

$$T = \frac{r \left(h + \frac{r}{3} \right) \rho g}{2}$$

Hence, surface tension of water can be determined.

In the actual experiment, three capillary tubes of different diameters are taken and the mean value of surface tension is calculated.

This method suffers from a number of drawbacks.

- (i) The angle of contact has been taken to be zero, because it cannot be measured inside the tube.
- (ii) The tube may not be of uniform area of cross-section.
- (iii) The tube may not be clean.

8.16 Examples of Capillarity

(1) A pen nib is split at the tip to provide the narrow capillary and the ink is drawn up to the tip continuously.

(2) In oil lamps and stoves, oil is drawn up through the capillaries of the wick.

(3) Pores in the blotting paper act as capillaries. When the blotting paper is kept in contact with ink, the ink is absorbed.

(4) Walls get damped in rainy season due to the absorption of water by the bricks by capillary action.